Chapter 2

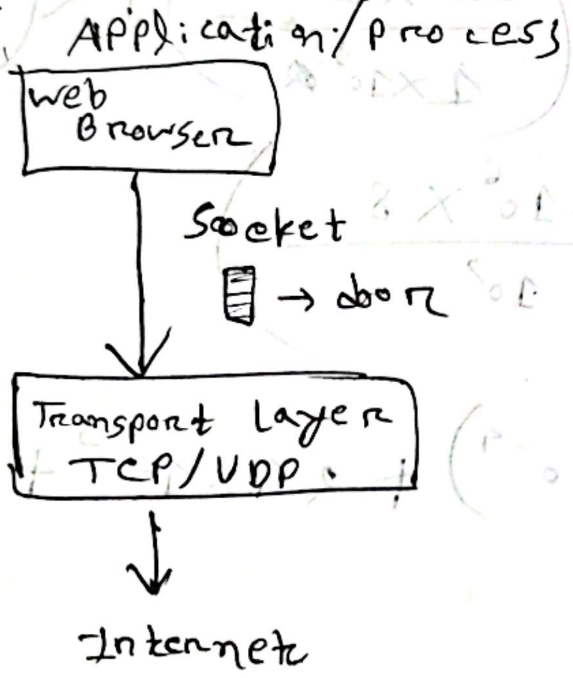
Client-server architecture

Server: Always-on

Permanent IP Address

Client: সবসময় online নাও থাকতে পারে

Socket:



[redacted]

Sockets

Application process and transport layer er moddhe door/interface.

Application tar message socket er maddhome transport layer e pathay.

Addressing Process

Ekta pc te ekta ip thake. But ekta pc te eksathe onkgulo kaj chole, ~~to~~ example.

- chrome
- Email
- Game

sudhu ip diye kon application-e data jab seta bojha jabe na.

IP address + port Number

HTTP Server:
Port 80

Mail Server:
Port 25

Application Layer Protocol

1. Message Types
2. Message syntax

- Message semantics
- rules
- open protocols
- proprietary protocols; Ex: Skype

TCP vs UDP

TCP	UDP
Reliable	Unreliable
Ex: Web, Email, File Transfer	Ex: Real Time Audio/Video
No Data loss,	

Non Persistent Vs Persistent

Non Persistent:

~~on~~ at most one object sent over one TCP connection

Persistent:

multiple objects can be sent over one single TCP connection.

RTT: time for a small packet to travel from client to server ~~and~~

Why web caching?

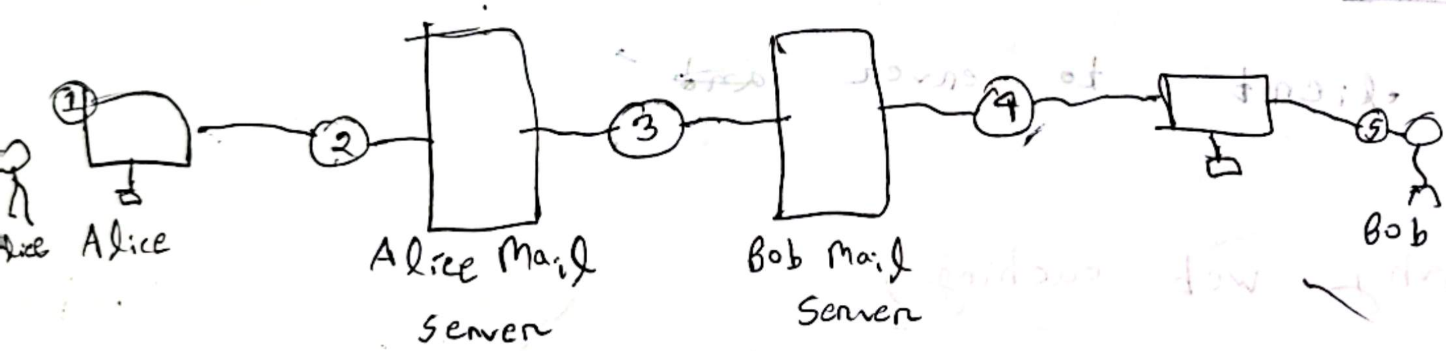
- Reduce Response Time
- Reduce Internet Traffic
- Access link er traffic kom e.

Electronic Mail

- User Agent
- Mail server
- SMTP

Mailservers

- Mailbox
- Message Queue



Mail Access Protocols

SMTP = Email deliver/send করার জন্য

POP/IMAP = email retrieve/read করার জন্য

POP3 = Authorization (Username/password দিয়ে user কে verify করে)

#

POP3 = Server-এ Email এর copy রেখে দেয়, যেন আমি device change করলেও সেটা হারিয়ে না যায়।

IMAP = সব message server এ রাখে

HTTP1 vs HTTP2 vs HTTP3

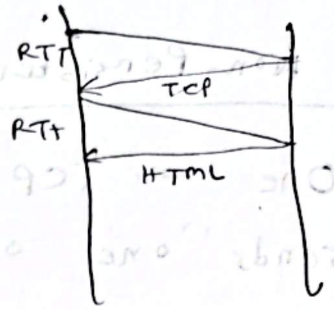
	HTTP1.1	HTTP2	HTTP3
Transport	TCP	TCP	QUIC over UDP
Main Idea	Basic	Faster	More Faster
Multiple Requests	Limited	Multiple request at the same time	Multiple request at the same time
Data Format	Text-Based	Binary Frames	Binary Frames
Performance	Comparatively slower	Faster than HTTP 1.1	Usually faster, especially on unreliable networks
Connection Setup	TCP	TCP	QUIC
Key Features	Persistent Connection	Multiplexing	QUIC + UDP + Multiplexing

Persistent vs Non-Persistent HTTP

Non-Persistent HTTP	Persistent HTTP
One ~~—~~ TCP connection sends one object	One TCP connection can send many objects.
After sending the object, the connection is closed.	The connection stays open
Many objects need many TCP connections	Many objects can used one TCP connection.
More connection overhead.	Less connection overhead.
Slower when there are many objects.	Faster for many objects

Example = Lecture _ No.2 - Application Layer.pptx

(ii)



For the base HTML file :

1 RTT → establish TCP connection

1 RTT → send HTTP request and respones

Initial Time = 2 RTT

(i) Non persistent without parallel.

Initial 2 RTT

For 30 images $= (30 \times 2)$; [30 image er jonno 30 ta TCP connection].

$= 60$

Total Time = 62 RTT

(Ans)

(ii) Non-persistent connection with 10 Parallel

10 টি করে image ekstahe download kora jabe.

Total images = 30

Number of rounds,

$$\frac{30}{10} = 3$$

Initial 2 RTT

For 3 round $= (3 \times 2)$ RTT [3 round er jonno 3 TCP

$= 6$ RTT

Total 8 RTT.

(iii) Persistent connection without Pipelining

Initial 2 RTT for TCP and HTML Request/Resp

30 টি Image এর jonno 30 RTT

Total = 30 + 2

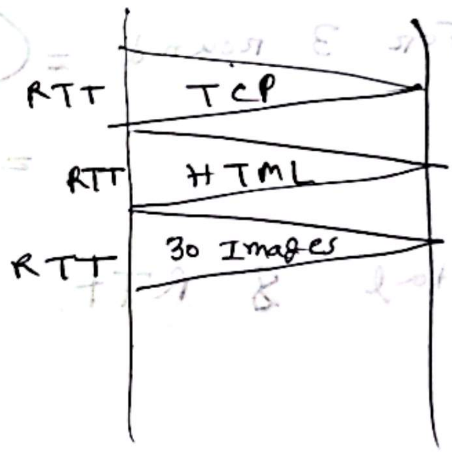
= 32 RTT

(iv) Persistent connection with pipe-lining:

Initial 2 RTT

Pipelining থাকার কারণে multiple image requests একসাথে/ পরপর পাঠানো যায়, প্রতিটির response - এর জন্য আলাদা করে wait করতে হয় না।

30 টা Image এর জন্য = 1 RTT



Total Time = 2 + 1

= 3 RTT

Shortcut for This Types Math

Excluding header and Trailer

শুধু IP(20) + TCP(20) বাদ যাবে

Including header and Trailer

DL (14+4) + IP(20) + TCP(20) সব বাদ যাবে

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1 Mbps = 1000 KBps = 10^6 bps

$\frac{10000}{10}$

1 Gbps = 10^9 bps

1 byte = 8 ~~byte~~ bits

1 msec = 10^{-3} sec

$$^{n}C_{r} \cdot p^{r} (1-p)^{n-r}$$

$$\sum_{i}^{n} {}^{n}C_{i} (p)^{i} (1-p)^{n-i}$$

$$\text{Transmission Time} = \frac{\text{Data size}}{\text{Throughput}}$$

Throughput = minimum transmission

Lap top → Wifi Router → ISP → Internet → Google Server

Nuts and bolts?

Protocol

Protocol 3 টা জিনিস define করে

(i) Message Format (Message dekhte kemon hobe)

(ii) Message order (কোন message আগে, কোনটা পরে

(iii) Action
Mesage পেলে কী করতে হবে

Client → Connection Request

Server → Response

Client → Connection Established

Forwarding and Routing

Forwarding

Packet আসার পর সেটিকে কোন output link দিয়ে পরের router এর দিকে পাঠাবে, seta fix করে।

Routing

source থেকে destination পর্যন্ত packet কোন route/ path দিয়ে যাবে সেটা determine করে।
পুরো internet এর best path খুঁজে বের করা

Circuit Switching:

when some call another one the whole line is reserved for himself. Noone can enter that line.

Source ——————— Destina

Reserved line

- Dedicated Bandwidth
- No sharing
- Predictable performance
- Idle অবস্থাতেও Resource Reserved

Packet Switching Vs Circuit Switching

Feature	Packet	Circuit
Resource	Shared	Dedicated
Path Setup	No	Yes (before communication)
Packet Loss	Possible	Very Low
Delay	Variable	Predictable
Bandwidth	On demand	Reserved
Best for	Internet, Web, Email	Traditional Phone Call

Internet structure: network of networks

Access Local ISP → Regional ISP → Global ISP

Example:

Home Network

↓

Access Local ISP

|

Regional ISP

|

Global ISP

Google

|

Another ISP (Google's Local ISP)

Layering Concept

ticket (purchase)	ticket (complain)
baggage (check)	baggage (claim)
gates (load)	gates unload
runway (takeoff)	runway landing
airplane routing	airplane routing

Advantages of Layering

- Simplifies complex systems
- Modular Design
- Easy maintenance
- Easy updating
- Independent implementation

OSI Model

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Link
1. Physical

TCP/IP Model

Application Layer

OSI এর 3 টি layer একসাথে

Application + Presentation + Session

Real Life Example

Suppose you open Youtube:

Unit

1. Application → HTTP request [Message]
2. Transport → TCP Data send. করে [Segmen
3. Network → IP address দিয়ে route করে [Packet]
4. Data Link → Frame বানায় [Frame]
5. Physical → Signal দিয়ে পাঠায় [Bit]

Transmission Control Protocol

TCP এর Main Features

1. Reliable
   - Data হারায় না

2. Connection - Oriented
   - আগে connection তৈরি হয় - তারপর data পাঠানো হয়

3. Ordered Delivery
   - Data যেভাবে পাঠানো হয়, সেভাবেই receive হয়

4. Error Detection

5. Flow Control
   - Receiver যতটুকু নিতে পারে, ততটুকুই sender পাঠায়

6. Congestion Control
   - Network Overload হলে speed কমায়

Throughput

Data পৌঁছানোর rate (sender → receiver

– একক সময়ে কত bits successfully transmit হয়

bps (bits per second)

Bottleneck

network এর সবচেয়ে slow (ধীর) অংশ, যেটা পুরো transmission এর speed কমিয়ে দেয়।

· System jotoi fast thakuk, ekta slow ~~part~~ puro system slow kore dey – setai bottleneck.

Layering Concept

Network system-কে কয়েকটা স্তরে ভাগ করা

- একটা complete system (internet) একসাথে বোঝা কঠিন
- তাই এটাকে ছোট ছোট অংশে ভাগ করা হয় = lay.

Real-Life Analogy (Air Travel)

1. Ticket (Booking)
2. Bagage check
3. Boarding
4. Flight

- প্রতিটা step আলাদা কাজ করে
- কিন্তু সব মিলে পুরো journey complete হয়

Packet Switching কীভাবে কাজ করে

1. Segmentation

Big message → small packet

2. Transmission

প্রতিটি message network এ পাঠানো হয়

3. Routing

Router:
- packet Receive করে
- header দেখে
- next hop ঠিক করে

4. Different Paths

- সব packet একই Path দিয়ে যাবে না
- আলাদা route নিতে পারে

5. Reassembly

- Receiver sob packet নিয়ে আবার origi
message বানায়